Joint SNR–EH Performance of Diversity Combining Techniques in LoRa Downlink Networks

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***Abstract*—To address the growing demand for energy-efficient wireless communications while maintaining satisfactory signal quality at the receiver, this paper presents a comprehensive investigation of the joint performance of signal reception and energy harvesting (EH) in LoRa downlink networks. The effects of key system parameters are analyzed, including the diversity combining schemes implemented at the gateway, namely transmit antenna selection (TAS), maximum ratio transmission (MRT), and equal gain transmission (EGT), as well as the signal-to-noise ratio (SNR) and the harvested energy thresholds at the end device (ED). Furthermore, closed-form expressions for the joint SNR–EH complementary cumulative distribution function (joint CCDF) are derived and presented. Monte Carlo simulations are also conducted to validate the accuracy of the theoretical analysis and to demonstrate the effectiveness of the proposed framework.**

***Index Terms*—Energy harvesting, LoRa downlink, TAS, MRT, EGT.**

I. Introduction

Numerous studies have been conducted to achieve both energy efficiency and reliable signal reception using different analytical frameworks and system models. Although existing works have investigated this problem under various scenarios, environments, and system configurations, several research gaps still remain and require further investigation. For instance, the study in [1] primarily focused on enhancing the reliability of long range wide area network (LoRaWAN) by proposing channel allocation and downlink operation mechanisms to improve the successful packet reception probability. However, the impact of energy harvesting (EH) and diversity combining techniques (DCTs) on the overall system performance was not considered. In [2], the authors investigated the use of transmit antenna selection and maximal ratio combining to improve both EH efficiency and communication reliability. Nevertheless, the joint relationship between signal-to-noise ratio (SNR) and EH performance was not addressed, particularly in long range (LoRa) networks. Furthermore, the work presented in [3] examined EH in LoRa networks with the objectives of improving throughput, maintaining user fairness, and reducing energy consumption. However, the effects of DTCs on the joint SNR–EH performance were not investigated.

According to a survey of recent studies, a considerable number of research works have investigated DTCs in wireless communication systems. For instance, the work in [4] focused on a simple optimal error analysis of selection combining (SC), and equal gain combining (EGC) over correlated Rayleigh fading channels. The error analysis developed for maximal ratio combining (MRC) can also be applied to both EGC and SC, since the signals at the outputs of the splitter are uncorrelated. A comprehensive review of MRC was presented in [5]. Furthermore, the study [6] investigated the application of SC and MRC in hybrid THz-RF systems and analyzed key performance metrics, including bit error rate, outage probability, and ergodic capacity, for emerging 6G networks.

Regarding energy harvesting, we focus on the simultaneous wireless information and power transfer (SWIPT) technology, which enables the use of the same radio frequency (RF) signal for both information transmission and energy supply. Instead of being considered only as interference, RF signals can be exploited as a valuable energy source. However, SWIPT introduces a trade-off between data transmission rate and harvested energy, which affects network performance. Although it improves energy efficiency and extending device lifetime, it requires careful design to balance communication and energy objectives [7]. Despite these benefits, EH faces several challenges, including limited and environment-dependent harvested energy, which makes stable operation difficult, especially in large-scale or high path-loss scenarios [8]. Additionally, in EH-enabled networks, increasing node density can increase interference and degrade performance (e.g., coverage probability), even though it may enhance EH opportunities [9]. Furthermore, for work [10], system performance strongly depends on practical EH models due to the non-linear behavior of harvesting circuits. The combination of limited energy, performance trade-offs, and network interference makes the system design and performance optimization significantly more complex.

Overall, existing studies have mainly focused on individual aspects of wireless communications and EH, while their joint performance has not been extensively investigated, particularly in LoRa downlink networks. As a representative low power wide area network (LPWAN) technology, LoRa has attracted significant attention for Internet of Things (IoT) applications due to its low power consumption, wide coverage, and low deployment cost [11], [12]. However, LoRa networks still face several challenges, including low data rates, reduced communication reliability at high spreading factors, imperfect orthogonality among spreading factors, and limited downlink resources caused by duty-cycle regulations and bandwidth limitations [13], [14]. These issues are further worsened by fading, interference, and packet collisions, especially in large-scale IoT networks. Although various resource management techniques, such as packet scheduling, spreading factor allocation, frequency assignment, and power control, have been proposed to improve LoRa performance [15], the joint SNR and EH performance analysis under various diversity combining schemes remains largely unexplored. This research gap motivates the present work, which aims to provide a unified analytical framework and derive closed-form expressions for evaluating the joint performance of signal reception and EH.

Motivated by the aforementioned observations and the potential benefits of advanced DCTs, this paper presents the performance of diversity reception schemes integrated with EH in LoRa downlink networks. The study aims to evaluate and compare the joint performance of information reception and EH, with the objective of improving communication reliability while maintaining effective harvested energy at the end device (ED). The principal contributions of this study are listed below:

(i) We investigate several key contributions of transmit antenna selection (TAS), maximum ratio transmission (MRT), and equal gain transmission (EGT) schemes integrated with SWIPT in LoRa downlink networks, including

- Performances of TAS, MRT, and EGT techniques are investigated and compared under different SNR and EH thresholds to determine which scheme provides the best and worst signal reception performance.
- The joint impact of SNR and EH thresholds on the performance of diversity combining schemes is investigated.
- The impacts of transmit power, transmission distance, and power-splitting ratio on the received signal quality are investigated. In particular, the impact of the power-splitting ratio on the joint SNR-EH performance is investigated, and the numerically best operating region is identified for the considered parameter settings.

(ii) A general closed-form expression is derived and verified by Monte Carlo simulations to evaluate the joint SNR and EH probability and analyze the effects of key system parameters on system performance, including threshold value, power-splitting ratio and transmit power.

II. System Parameter Analysis

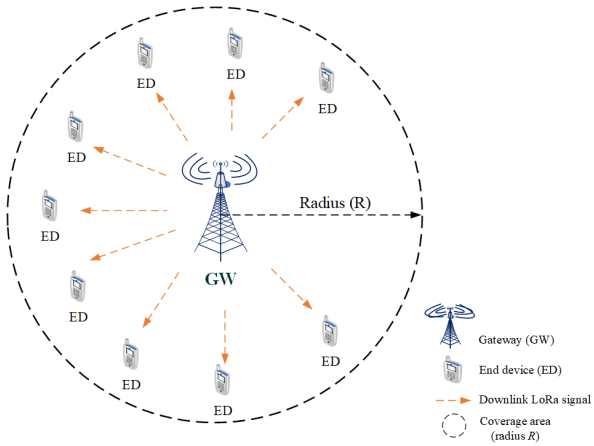


Fig. 1. System model of a conventional LoRa downlink network.

Consider a downlink LoRa network consisting of a multi-antenna gateway and a single-antenna end device, where the wireless channel is modeled as Rayleigh fading. At receiver side, ED is equipped with a SWIPT architecture, enabling both information decoding and EH from the received signal. At transmitter side, employs transmit diversity combining techniques, including TAS, MRT, and EGT, to improve communication reliability and system performance.

As illustrated in Fig.1, the considered network follows a circular topology, where gateway is located at the center of a circular area with radius (R), while ED is randomly distributed within the coverage region to receive the signals transmitted by the gateway.

A. Channel modeling

It is assumed that wireless channel between gateway and ED experiences path loss and Rayleigh fading only, whereas the shadowing effect is ignored throughout the analysis.

In this model, the small-scale fading channel coefficient h_i is modeled as a circularly symmetric complex Gaussian random variable with zero mean and variance Ω, i.e., $h_i \sim CN(0,\Omega)$. Consequently, the channel power gain $|h_i|^2$ follows an exponential distribution with mean Ω, i.e., $|h_i|^2 \sim Exp(\Omega)$.

The large-scale path loss can be formulated as follows [16]:

$$L\{r_{\{GW->ED\}}\} = L_0 r^{\delta}_{\{GW->ED\}}, \tag{1}$$

where L_0 denotes the path loss constant at the reference distance; $r_{GW->ED}$ represents the transmission distance between gateway and ED. Here, $\delta \in (2,8)$ is the path loss exponent.

B. Spreading factor

The spreading factor (SF) is allocated according to the distance $r_{GW->ED}$ between gateway and ED. Specifically, SF is selected from the set $\{SF7,...,SF12\}$ using a distance-based allocation scheme. Let SF_ω denote the SF assigned to $\omega - th$ region, where $\omega \in \{7,...,12\}$. Assuming that the gateway is located at the center of a circular coverage area with R, the coverage area is divided into six concentric rings, each with a radial width of $\frac{R}{6}$. Therefore, SF_ω is assigned when

$$\frac{(\omega-7)\,R}{6} < r_{GW->ED} \leq \frac{(\omega-6)\,R}{6}. \tag{2}$$

Here, although EDs are randomly distributed within the coverage region, the analytical evaluation for each SF_ω is performed using the worst-case distance associated with that SF region. Specifically, the representative distance for SF_ω chosen as the outer boundary of the corresponding distance interval.

C. Diversity combining techniques at the gateway

To combat fading and enhance signal quality, several DCTs have been proposed for wireless communication systems, among which TAS, MRT, and EGT are the most widely used.

TAS is a diversity technique that selects the transmit antenna with the highest channel gain toward ED. Consequently, the effective channel power gain is given by

$$|h|^2_{TAS} = \max_{i=1,...,M} |h_i|^2. \tag{3}$$

In MRT scheme, signals from all diversity branches are combined with weights optimized according to the channel coefficients. As a result, the output SNR is maximized, and the equivalent channel power gain is expressed as

$$|h|^2_{MRT} = \sum_{i=1}^{M} |h_i|^2. \tag{4}$$

Regarding EGT technique, the phases of all diversity branches are coherently aligned, and common gain is applied to each branch without requiring accurate channel amplitude estimation. Accordingly,

$$|h|^2_{EGT} = \left(\sum_{i=1}^{M} |h_i|\right)^2 \Big/ M. \tag{5}$$

D. Received signal

The considered SWIPT receiver adopts a conventional power-splitting architecture. The received signal is divided into two parts, including one part is used for information decoding, while the other part is used for energy harvesting. Following the common assumption in SWIPT systems, the main receiver noise is introduced in the information-decoding circuit after the power splitter. Therefore, the power-splitting factor φ affects only the useful received signal power, while the noise power remains unchanged. Based on this assumption, the SNR at the ED using the spreading factor SF_ω during the $\varepsilon - th$ time slot is given by:

$$\xi^{\varepsilon}_{\omega_\theta} = \frac{\varphi P_{tr}\,|h|^2_\theta}{L\{r_{\{GW->ED,\omega\}}\}\sigma^2} = \frac{\varphi P_{tr}\,|h|^2_\theta}{L_0 r^{\delta}_{\{GW->ED,\omega\}}\sigma^2}. \tag{6}$$

where $\theta \in \{TAS, MRT, EGT\}$. It should be noted that, if the antenna noise before the power splitter is taken into account, the SNR would depend on both the noise before and after the power splitter. In this work, we consider only the post-splitting receiver noise.

By setting $\gamma = \frac{\varphi P_{tr}}{L_0 r^{\delta}_{\{GW->ED,\omega\}}\sigma^2}$, where φ denotes the SWIPT power-splitting factor, where a portion φ of the received signal power is used for information decoding, and the remaining portion $(1-\varphi)$ is allocated to EH, P_{tr} is the gateway transmit power, $L\{r_{\{GW->ED,\omega\}}\}$ represents the large-scale path loss between the gateway and the worst-case ED operating with SF_ω, where $r_{\{GW->ED,\omega\}}$ denotes the worst-case distance associated with the region assigned to SF_ω. Specifically, for SF_ω , the worst-case distance is selected as the outer boundary of its corresponding distance interval defined in (2). Consequently, the worst-case distance equals $\frac{(\omega-6)R}{6}$, while the overall cell-edge distance becomes R for SF_{12}. And $\sigma^2 = 10^{(-174+NF+10\log_{10}(BW))/10}$ $[mW]$ denotes the additive white Gaussian noise (AWGN), which is modeled as a zero-mean Gaussian noise with a constant power spectral density over the frequency band, while BW [Hz] and NF [dB] denote the transmission bandwidth and the noise figure, respectively. Here, the effective channel gain $|h|^2_\theta$ varies depending on the considered diversity combining technique. According, the received signal is expressed as follows:

TAS	**MRT**	**EGT**
$\xi^{\varepsilon}_{\omega_TAS} = \gamma\,\|h\|^2_{TAS}$	$\xi^{\varepsilon}_{\omega_MRT} = \gamma\,\|h\|^2_{MRT}$	$\xi^{\varepsilon}_{\omega_EGT} = \gamma\,\|h\|^2_{EGT}$

E. Energy harvesting

Assuming that the contribution of noise to the harvested energy is insignificant, the noise component is neglected. Thus, the general expression for the harvested energy at DCTs is given by

$$\mathrm{E}_H^{\theta} = (1-\varphi)\rho P_r T = \frac{(1-\varphi)\rho P_{tr} T}{L_0 r^{\delta}_{\{GW->ED,\omega\}}} |h|^2_{\theta}. \quad (7)$$

By setting $K = \frac{(1-\varphi)\rho P_{tr} T}{L_0 r^{\delta}_{\{GW->ED,\omega\}}}$, where ρ is the energy conversion efficiency $(0 < \rho < 1)$, T is the energy harvesting time factor, and $(1-\varphi)$ is the fraction of the received energy used for EH. The corresponding DCTs are considered in the following cases:

TAS	MRT	EGT
$\mathrm{E}_H^{TAS} = K\,\lvert h\rvert^2_{TAS}$	$\mathrm{E}_H^{MRT} = K\,\lvert h\rvert^2_{MRT}$	$\mathrm{E}_H^{EGT} = K\,\lvert h\rvert^2_{EGT}$

III. Performance analysis

The joint probability of SNR and EH is given by

$$\mathrm{P}^{\theta}_{SE}(\omega) = \Pr\left(\xi^{\varepsilon}_{\omega_\theta} \geq \xi_{th}, \mathrm{E}^{\theta}_H \geq \mathrm{E}_{th}\right), \quad (8)$$

where $\Pr\{.\}$ denotes the probability operator, E_{th} denotes EH threshold, and $\xi_{th(\omega)} = \{-6, -9, -12, -15, -17.5, -20\}$ $[dB]$ denotes SNR threshold for the $\omega - th$ spreading factor, which is specified by the relevant technical standards [17]. Replacing $\xi^{\varepsilon}_{\omega_\theta}$ and E^{θ}_H in the previous expression (8), we obtain [18]

$$\mathrm{P}^{\theta}_{SE}(\omega) = \Pr\left(|h|^2_{\theta} \geq \frac{\xi_{th}}{\gamma}, |h|^2_{\theta} \geq \frac{\mathrm{E}_{th}}{K}\right). \quad (9)$$

The joint SNR–EH probabilities corresponding to TAS, MRT, and EGT are derived as follows:

1) For TAS case: The combining rule is $|h|^2_{TAS} \geq \max\left(\frac{\xi_{th}}{\gamma}, \frac{\mathrm{E}_{th}}{K}\right)$, and the probability can be expressed as

$$\mathrm{P}^{TAS}_{SE}(\omega) = \Pr\left(\max_{i=1,\ldots,M} |h_i|^2 \geq \max\left(\frac{\xi_{th}}{\gamma}, \frac{\mathrm{E}_{th}}{K}\right)\right). \quad (10)$$

Let $x_0 = \max\left(\frac{\xi_{th}}{\gamma}, \frac{\mathrm{E}_{th}}{K}\right)$. Then, $\mathrm{P}^{TAS}_{SE}(\omega) = \Pr\left(|h|^2_{TAS} \geq x_0\right)$ can be obtained. Consider M independent and identically distributed (i.i.d.) exponentially distributed random variables. Since $|h_i|^2 \sim Exp(\Omega)$ follows an exponential distribution, the distribution of $|h|^2_{TAS} = \max_{i=1,\ldots,M} |h_i|^2$ is given as follows:

- Cumulative distribution function (CDF)

$\mathrm{P}_{|h|^2_{TAS}}(x_0) = \Pr\left(|h|^2_{TAS} < x_0\right) = \left(1 - e^{-x_0/\Omega}\right)^M$.

- Complementary cumulative distribution function (CCDF)

$\overline{\mathrm{P}}_{|h|^2_{TAS}}(x_0) = \Pr\left(|h|^2_{TAS} \geq x_0\right) = 1 - \left(1 - e^{-x_0/\Omega}\right)^M$.

Therefore, P^{TAS}_{SE} is determined from CCDF of the maximum among the exponentially distributed random variables, given by

$$\mathrm{P}^{TAS}_{SE}(\omega) = 1 - \left(1 - e^{-\max\left(\frac{\xi_{th}}{\gamma}, \frac{\mathrm{E}_{th}}{K}\right)/\Omega}\right)^M. \quad (11)$$

2) In the case of MRT: Similar to the TAS case, $|h|^2_{MRT} = \sum_{i=1}^{M} |h_i|^2$ follows an exponential distribution. The CCDF of the corresponding Gamma-distributed random variable is given by

$$\begin{aligned}\mathrm{P}^{MRT}_{SE}(\omega) &= \mathrm{P}_{|h|^2_{MRT}}(x_0) = \Pr\left(|h|^2_{MRT} \geq x_0\right) \\ &= 1 - \left(1 - e^{-x_0/\Omega} \sum_{k=0}^{M-1} \frac{(x_0/\Omega)^k}{k!}\right) \\ &= 1 - \left(1 - e^{-\max\left(\frac{\xi_{th}}{\gamma}, \frac{\mathrm{E}_{th}}{K}\right)/\Omega} \sum_{k=0}^{M-1} \frac{\left(\max\left(\frac{\xi_{th}}{\gamma}, \frac{\mathrm{E}_{th}}{K}\right)/\Omega\right)^k}{k!}\right)\end{aligned} \quad (12)$$

3) Considering EGT scheme: As in TAS and MRT cases, the expression P^{EGT}_{SE} can be written as

$$\begin{aligned}\mathrm{P}^{EGT}_{SE}(\omega) &= \Pr\left(|h|^2_{EGT} \geq x_0\right) \\ &= \Pr\left(\sum_{i=1}^{M} |h_i| \geq \sqrt{M x_0}\right) \\ &\approx Q\left(\frac{\sqrt{\frac{\max\left(\frac{\xi_{th}}{\gamma}, \frac{\mathrm{E}_{th}}{K}\right)}{\Omega}} - \frac{\sqrt{\pi M}}{2}}{\sqrt{1 - \frac{\pi}{4}}}\right)\end{aligned}. \quad (13)$$

where Q[.] denotes Gaussian Q-function, which represents the right-tail probability of a standard normal distribution.

IV. Simulation Results

To validate the analytical results, Monte Carlo simulations are carried out using parameter settings listed below to evaluate system performance: $SF = \{7, 8, \ldots, 12\}$, $BW = 125$ KHz, $NF = 8$ dB, $f_c = 868$ MHz, $\xi_{th} = \{-6, -9, -12, -15, -17.5, -20\}$ dB and $\delta = 2.95$. Furthermore, effects of varying the following parameters are examined in this paper, including E_{th}, φ, P_{tr}, và r. Unless otherwise stated, these parameters are set to fixed values as follows: $\mathrm{E}_{th} = 10^{-14} J$, $\varphi = 0.7$, $P_{tr} = 14$ dBm, và $R = 6$ km.

Fig. 2 investigates the impact of E_{th} while keeping ξ_{th} fixed. It can be observed that joint CCDF varies across different SFs. However, it consistently decreases as E_{th} increases. In addition, the performance ranking of the diversity combining techniques remains unchanged, with TAS providing the poorest performance, MRT achieving the best performance, and EGT achieving performance between those of TAS and MRT. Specifically, at SF = 8 and $\mathrm{E}_{th} = 0(J)$, the joint CCDF achieved by TAS approaches 0.2, while MRT attains approximately 0.4 and EGT falls between 0.2 and 0.4. When $\mathrm{E}_{th} = 5.10^{-14}(J)$, the joint CCDF of all combining schemes approaches 0.

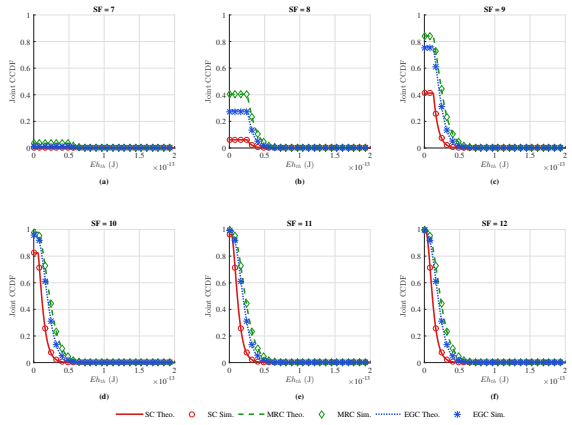


Fig. 2. Impact of E_{th} on joint CCDF of successful information decoding and energy harvesting. Subfigures (a), (b), (c), (d), (e), and (f) correspond to SF = 7, 8, 9, 10, 11 and 12, respectively. Lines represent the analytical results (Theo.) derived from (11), (12), and (13), whereas markers indicate the simulation results (Sim.) obtained through Monte Carlo simulations.

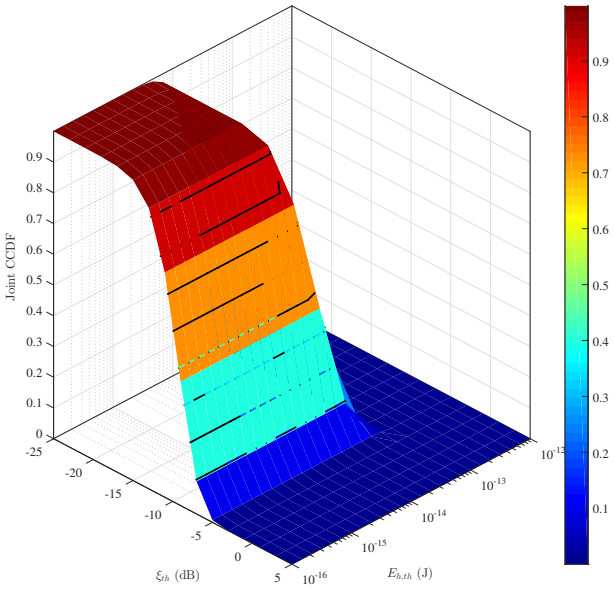


Fig. 3. Impact of $(\xi_{th}, \mathrm{E}_{th})$ on the joint CCDF of successful information decoding and energy harvesting. The surface color reflects the magnitude of the joint CCDF, with warmer colors corresponding to higher values and cooler colors indicating lower values.

Fig. 3 illustrates the variation of the joint CCDF when both SNR and EH thresholds are simultaneously varied. It can be observed that the joint CCDF increases as either ξ_{th} or E_{th} decreases, i.e., increasing the probability of simultaneously achieving successful signal decoding and energy harvesting. Specifically, the red region, corresponding to the joint CCDF remains around 0.9 when ξ_{th} is very low (approximately from $5\,\mathrm{dB}$ to $-5\,\mathrm{dB}$) and E_{th} is also very small (approximately from 10^{-12} to $10^{-13}(J)$). In contrast, when ξ_{th} increases to the range from $-25\,\mathrm{dB}$ to $-15\,\mathrm{dB}$ or E_{th} increases from 10^{-16} to $10^{-15}(J)$, the joint CCDF declines to around 0.1. This indicates that strict SNR and EH requirements considerably reduce the probability of simultaneously achieving reliable communication and sufficient energy harvesting.

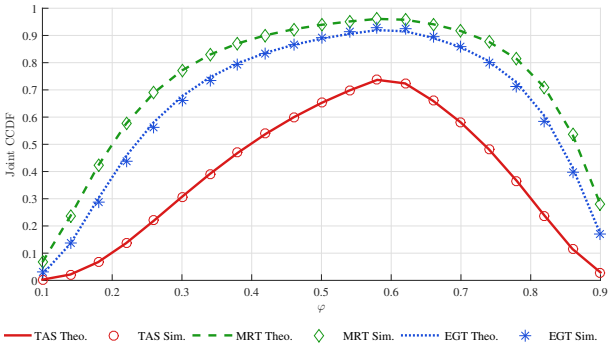


Fig. 4. Impact of the energy-splitting factor for information decoding (φ) on joint CCDF of successful information decoding and energy harvesting. Lines represent the analytical results (Theo.) derived from (11), (12), and (13), whereas markers indicate the simulation results (Sim.) obtained through Monte Carlo simulations.

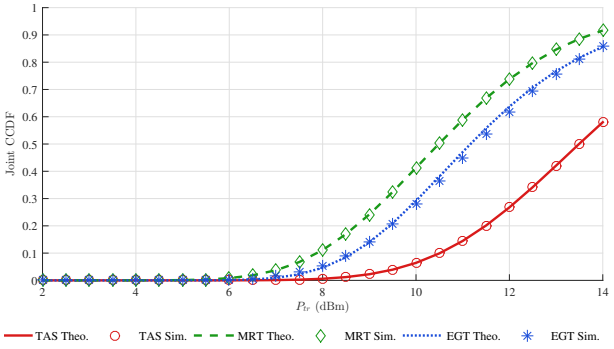


Fig. 5. Impact of the gateway transmit power (P_{tr}) on joint CCDF of successful information decoding and energy harvesting. Lines represent the analytical results (Theo.) derived from (11), (12), and (13), whereas markers indicate the simulation results (Sim.) obtained through Monte Carlo simulations.

Fig. 4 presents φ on the joint CCDF. It can be observed that the joint CCDF increases as rises from 0.1 to 0.6, and then gradually decreases as φ further increases from 0.6 to 0.9. This trend reveals the trade-off between information decoding and EH in the SWIPT receiver. Furthermore, under the adopted SWIPT receiver noise model, the highest joint CCDF is achieved at approximately $\varphi = 0.6$. In this case, TAS exhibits the worst performance, whereas MRT onsistently achieves the best performance. Specifically, the joint CCDF of TAS reaches approximately 0.7, while those of EGT and MRT are around 0.9–1, with MRT achieving the highest value and approaching 1.

Fig. 5 shows that the joint CCDF increases steadily with the transmit power. As P_{tr} increases, both the received signal quality and the harvested energy improve, leading to a higher probability of SNR and EH. Moreover, the strong agreement between the theoretical and simulation results confirms the validity of the analytical expression derived in (11), (12), and (13). For instance, at $P_{tr} = 10dBm$, the joint CCDF values

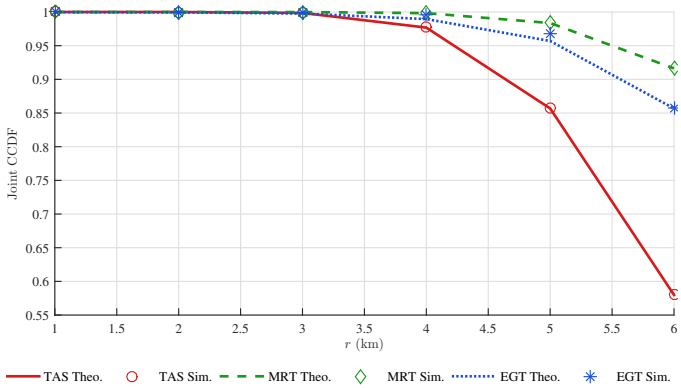


Fig. 6. Impact of the transmission distance between gateway and end device (r) on the joint CCDF of successful information decoding and energy harvesting. Lines represent the analytical results (Theo.) derived from (11), (12), and (13), whereas markers indicate the simulation results (Sim.) obtained through Monte Carlo simulations.

of TAS, EGT, and MRT are approximately below 0.1, 0.3, and 0.4, respectively, highlighting the better performance of MRT.

As illustrated in Fig. 6 joint CCDF decreases with increasing transmission distance r from 1 km to 6 km. This reduction is mainly caused by the increased path loss, which reduces both received SNR and harvested energy. Among the considered diversity combining schemes, TAS reveals the fastest performance reduction, with the joint CCDF dropping from approximately 1 at $r = 3km$ to around 0.55–0.60 at $r = 6km$. Meanwhile, MRT consistently provides the highest joint CCDF across all transmission distances, reaching approximately 0.95 at $r = 6km$, demonstrating better robustness path-loss and fading as the transmission distance increases.

V. Conclusions

In this study, closed-form expressions for P_{SE}^{θ} were derived and analyzed for diversity combining schemes integrated with SWIPT in a LoRa downlink network. Monte Carlo simulation results validated the accuracy of the derived analytical expressions and provided performance evaluations. The results demonstrated that MRT achieves the highest joint CCDF among all considered schemes, while the system performance is significantly influenced by SNR and EH thresholds, transmit power, transmission distance, and power-splitting ratio. Moreover, numerical results reveal that the best operating power-splitting region depends on the considered system configuration and reflects the trade-off between information decoding and energy harvesting. The proposed framework can be extended in several research directions. For instance, future studies may focus on more realistic fading environments and imperfect channel state information (CSI) conditions [19], [20].

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